RISE Internship
Data Science & Analytics
Project List

Tools:

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn
- Jupyter Notebook

Project 1: Industry-Oriented Customer Behavior Analysis for Business Decision Making

Problem Statement:

Companies collect large volumes of customer data but struggle to convert it into actionable insights. Without proper analysis, businesses fail to understand customer behavior, preferences, and churn risk, leading to poor decision-making and revenue loss.

Objective:

To analyze real-world customer data and generate insights that help businesses improve customer retention, targeting, and overall strategy.

Requirements:

- Customer dataset ingestion
- Data cleaning and preprocessing
- Exploratory data analysis
- Customer segmentation analysis
- Identification of behavior patterns
- Basic predictive analysis for churn or purchase intent
- Visualization of key insights
- Business-focused insight report

Expected Outcome:

- Hands-on experience with real business datasets
- Strong data analytics portfolio project
- Understanding of data-driven decision making
- Job relevance for Data Analyst, Business Analyst, Junior Data Scientist roles

Tools:

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn
- Jupyter Notebook

Project 2: Industry-Oriented Sales Forecasting and Revenue Prediction System

Problem Statement:
Businesses need accurate sales forecasts to plan inventory, marketing, and revenue targets. Manual forecasting or intuition-based planning often leads to overstocking or missed revenue opportunities. Data-driven forecasting is critical for business growth.

Objective:
To build a data science solution that analyzes historical sales data and predicts future sales trends and revenue.

Requirements:

- Historical sales dataset ingestion
- Data preprocessing and feature engineering
- Trend and seasonality analysis
- Time-based data visualization
- Sales forecasting using machine learning models
- Model performance evaluation
- Forecast visualization and interpretation
- Business-oriented insights report

Expected Outcome:

- Practical exposure to forecasting and predictive analytics
- Strong industry-relevant data science project
- Understanding of demand planning and revenue analytics
- Job relevance for Data Analyst, Data Scientist, Business Intelligence Analyst roles

Tools:

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn
- Jupyter Notebook

Project 3: Industry-Oriented Fraud Detection Analysis Using Data Analytics

Problem Statement:

Banks, fintech companies, and e-commerce platforms face increasing fraud transactions that cause financial losses and damage customer trust. Manual fraud detection is inefficient and slow. Data-driven analysis is required to identify suspicious patterns and reduce fraud risk.

Objective:

To analyze transaction data and build a data analytics solution that identifies potentially fraudulent activities based on patterns and anomalies.

Requirements:

- Transaction dataset ingestion
- Data cleaning and preprocessing
- Exploratory analysis of transaction behavior
- Identification of anomaly patterns
- Feature engineering for fraud indicators

RISE Internship
Data Science & Analytics
Project List

- Fraud classification using analytics or ML models
- Evaluation of detection performance
- Visualization of fraud trends and insights

Expected Outcome:

- Hands-on experience with fraud analytics use cases
- Strong industry-relevant data science portfolio project
- Understanding of risk and anomaly detection
- Job relevance for Data Analyst, Fraud Analyst, Data Scientist roles

Tools:

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn
- Jupyter Notebook

Project 4: Industry-Oriented HR Analytics for Employee Attrition Prediction

Problem Statement:
Organizations face high costs due to employee attrition and lack visibility into the factors driving resignations. HR teams need data-driven insights to predict attrition risk and improve employee retention strategies.

Objective:
To analyze HR data and predict employee attrition using data science techniques to support strategic HR decision-making.

Requirements:

- HR dataset ingestion
- Data cleaning and preprocessing
- Exploratory analysis of employee attributes
- Feature engineering for attrition factors
- Attrition prediction using analytics or ML models
- Model performance evaluation
- Visualization of attrition drivers
- HR-focused insights and recommendationsocumentation

Expected Outcome:

- Hands-on experience with HR analytics
- Strong industry-relevant data science project
- Understanding of workforce analytics
- Job relevance for Data Analyst, People Analytics Analyst, Data Scientist roles

Tools:

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn
- Jupyter Notebook

Project 5: Industry-Oriented Marketing Campaign Performance Analysis

Problem Statement:

Companies invest heavily in digital marketing campaigns but often lack clear insights into which campaigns actually drive conversions and revenue. Without proper data analysis, marketing budgets are wasted and ROI remains unclear.

Objective:

To analyze marketing campaign data and measure campaign effectiveness to support data-driven marketing decisions.

Requirements:

- Marketing campaign dataset ingestion
- Data cleaning and preprocessing
- Analysis of campaign reach and engagement
- Conversion and ROI analysis
- Customer response behavior analysis

- Comparison of multiple campaign performances
- Visualization of key marketing metrics
- Actionable insights and recommendations

Expected Outcome:

- Hands-on experience with marketing analytics
- Strong industry-relevant data science project
- Understanding of ROI and performance metrics
- Job relevance for Data Analyst, Marketing Analyst, Business Analyst roles

Tools:

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn
- Jupyter Notebook

Project 6: Industry-Oriented Supply Chain and Inventory Demand Analysis

Problem Statement:
Manufacturing and retail companies struggle with inventory imbalance due to poor demand estimation. Overstocking increases holding costs, while understocking leads to lost sales. Data-driven supply chain analytics is essential to optimize inventory decisions.

Objective:
To analyze supply chain and inventory data to identify demand patterns and support optimal inventory planning.

Requirements:

- Inventory and sales dataset ingestion
- Data preprocessing and validation
- Demand trend and seasonality analysis
- Inventory turnover analysis
- Stock-out and overstock pattern identification

- Basic demand prediction analysis
- Visualization of inventory metrics
- Supply chain optimization insights

Expected Outcome:

- Hands-on experience with social engineering techniques
- Understanding of human-factor security risks
- Industry-ready cybersecurity portfolio project
- Job relevance for Ethical Hacker, Security Awareness Analyst, SOC Analyst roles

Tools:

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn
- Jupyter Notebook

Project 7: Industry-Oriented Financial Data Analysis and Risk Assessment

Problem Statement:
Financial institutions must assess risk in investments, loans, and portfolios to avoid losses and comply with regulations. Manual analysis is insufficient for large financial datasets. Data-driven financial analytics is essential for informed decision-making.

Objective:
To analyze financial data and assess risk patterns to support investment and credit decision-making.

Requirements:

- Financial dataset ingestion
- Data cleaning and preprocessing
- Exploratory financial data analysis
- Risk indicator identification
- Trend and volatility analysis

- Basic risk classification or scoring
- Visualization of financial insights
- Financial risk assessment report

Expected Outcome:

- Hands-on experience with financial analytics
- Strong industry-relevant data science project
- Understanding of risk and financial metrics
- Job relevance for Data Analyst, Financial Analyst, Risk Analyst roles

Tools:

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn
- Jupyter Notebook

Project 8: Industry-Ready End-to-End Business Intelligence and Analytics Dashboard

Problem Statement:
Organizations collect data across departments but lack a unified view for decision-making. Without integrated analytics dashboards, leaders struggle to track KPIs, identify trends, and respond quickly to business changes.

Objective:
To build an end-to-end analytics solution that transforms raw business data into an interactive dashboard with actionable insights.

Requirements:

- Multi-source business data ingestion
- Data cleaning and transformation
- Key KPI identification
- Analytical metric computation
- Interactive dashboard creation

- Trend and performance analysis
- Business insight generation
- Final presentation and documentation

Expected Outcome:

- End-to-end experience in data analytics workflows
- Strong industry-grade data science capstone project
- Understanding of business intelligence concepts
- Job relevance for Data Analyst, Business Intelligence Analyst, Data Scientist roles